

CASE STUDY 101

LearnHub

Database Management System

PostgreSQL Database Project Documentation

Student Name: Aditya Yadav
Cohort: Sam Altman
Roll Number: 150096725121

Technology: PostgreSQL 18.4
Database: learnhub_db
Environment: VS Code + PostgreSQL

1. Introduction

LearnHub is a relational database system designed to manage students, instructors, courses, enrollments and academic assessments. The project demonstrates the practical use of PostgreSQL for database creation, table design, constraints, relationships, sample data insertion, SQL querying, aggregation, joins, subqueries and database views.

The database is structured around five main entities: Student, Instructor, Course, Enrollment and Assessment. These entities are connected through primary-key and foreign-key relationships to maintain data consistency.

2. Objectives

- Design a relational database for an academic learning platform.
- Create tables using appropriate primary keys, foreign keys, constraints, and data types.
- Insert realistic sample data into the database.
- Perform SQL queries using SELECT, JOIN, WHERE, GROUP BY, ORDER BY, aggregate functions, and subqueries.
- Create a database view for an instructor dashboard.
- Demonstrate referential integrity and validation using SQL constraints.
- Document the complete database implementation and execution results.

3. Problem Statement

Educational platforms need a structured way to store and retrieve information about students, instructors, courses, enrollments, and assessments. Managing these records independently can lead to duplicate data, inconsistent records, and difficulty in generating useful academic reports. LearnHub addresses this requirement through a normalized relational database implemented in PostgreSQL.

4. System Overview

The LearnHub database follows a relational model. Students enroll in courses, courses are assigned to instructors, and each enrollment can have multiple assessments. The InstructorDashboard view combines these relationships to provide enrollment counts and average assessment scores for each course.

5. Technologies Used

Technology	**Details**
Database Management System	PostgreSQL 18.4
SQL Client / IDE	Visual Studio Code
SQL Extension	PostgreSQL extension for VS Code
Operating Environment	macOS
Database Name	learnhub_db

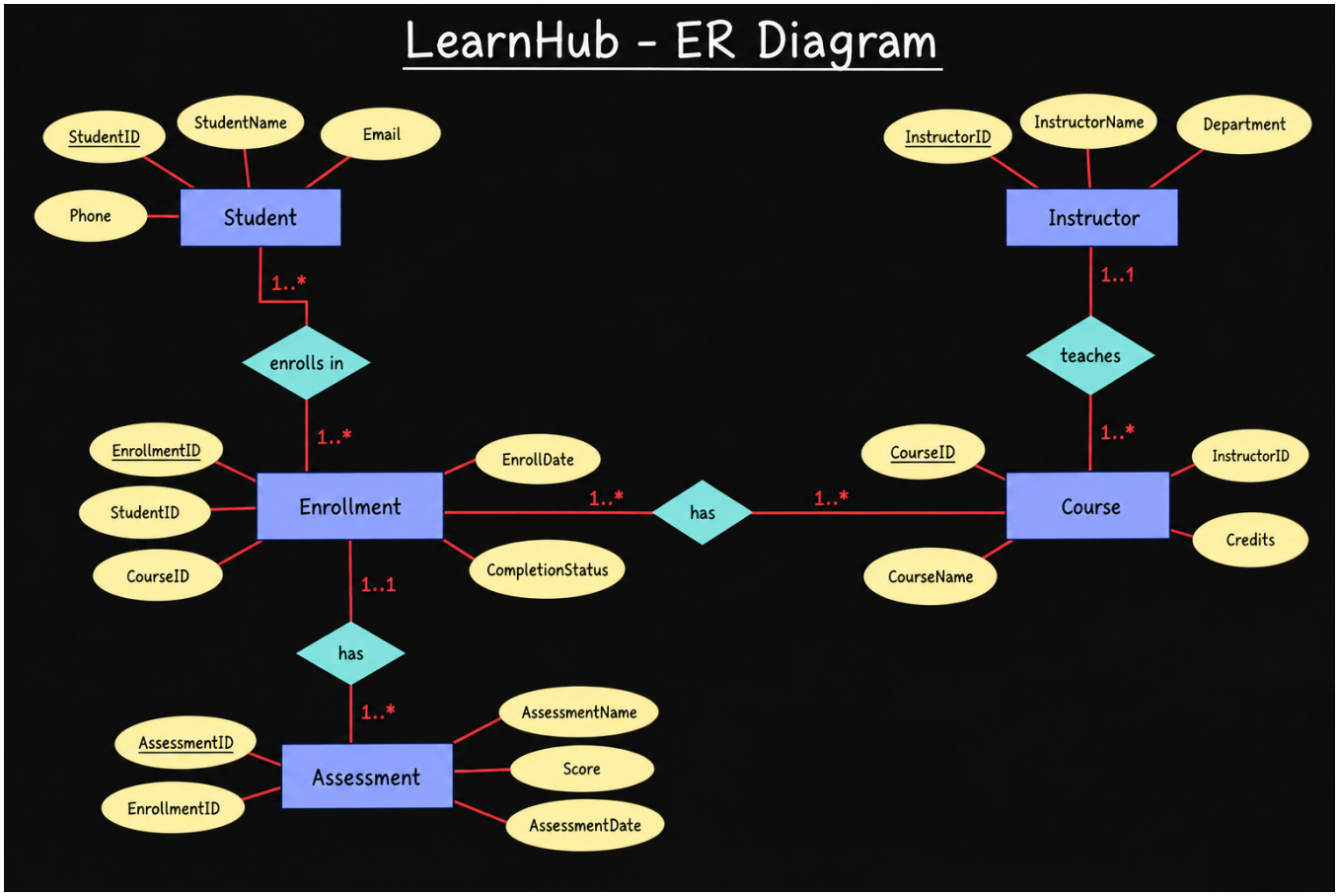
6. Database Design

The database contains five primary tables. Each table represents a major entity in the LearnHub system.

Table	Primary Key	Important Attributes
Student	StudentID	StudentName, Email, Phone
Instructor	InstructorID	InstructorName, Department
Course	CourseID	CourseName, InstructorID, Credits
Enrollment	EnrollmentID	StudentID, CourseID, EnrollDate, CompletionStatus
Assessment	AssessmentID	EnrollmentID, AssessmentName, Score, AssessmentDate

7. Entity Relationship Diagram

The ER diagram represents the entities, attributes, relationships, and cardinalities used in the LearnHub database.



Main relationships: Student enrolls in courses through Enrollment; Instructor teaches courses and Enrollment has Assessment records.

8. Table Structure and Constraints

Student

Column	Data Type	Constraint / Purpose
StudentID	INT	Primary Key
StudentName	VARCHAR(100)	NOT NULL
Email	VARCHAR(100)	NOT NULL, UNIQUE
Phone	VARCHAR(15)	NOT NULL

Instructor

Column	Data Type	Constraint / Purpose
InstructorID	INT	Primary Key
InstructorName	VARCHAR(100)	NOT NULL
Department	VARCHAR(100)	NOT NULL

Course

Column	Data Type	Constraint / Purpose
CourseID	INT	Primary Key
CourseName	VARCHAR(100)	NOT NULL
InstructorID	INT	NOT NULL, Foreign Key
Credits	INT	NOT NULL, CHECK (Credits > 0)

Enrollment

Column	Data Type	Constraint / Purpose
EnrollmentID	INT	Primary Key
StudentID	INT	NOT NULL, Foreign Key
CourseID	INT	NOT NULL, Foreign Key
EnrollDate	DATE	NOT NULL
CompletionStatus	VARCHAR(20)	NOT NULL, CHECK
StudentID + CourseID	Composite	UNIQUE

Assessment

Column	Data Type	Constraint / Purpose
AssessmentID	INT	Primary Key
EnrollmentID	INT	NOT NULL, Foreign Key
AssessmentName	VARCHAR(100)	NOT NULL
Score	DECIMAL(5,2)	NOT NULL, CHECK 0–100
AssessmentDate	DATE	NOT NULL

9. Sample Data

Table	Records Inserted
Student	8
Instructor	3
Course	6
Enrollment	13
Assessment	20

The sample data includes students, instructors, courses, course enrollments, and assessment results. The data was inserted using the 03_insert_data.sql script.

	instructorid	instructorname	department
1	101	Dr. Neha Sharma	Computer Science
2	102	Prof. Rahul Patil	Information Technology
3	103	Dr. Sneha Kulkarni	Artificial Intelligence

	courseid	coursename	instructorid	credits
1	201	Database Management Systems	101	4
2	202	Computer Networks	102	4
3	203	Java Programming	101	3
4	204	Artificial Intelligence	103	4
5	205	Web Development	102	3
6	206	Data Structures	103	4

	enrollmentid	studentid	courseid	enrolldate	completionstatus
1	301	1	201	2026-07-01	Completed
2	302	2	201	2026-07-02	Completed
3	303	3	201	2026-07-03	Enrolled
4	304	4	202	2026-07-04	Completed
5	305	5	202	2026-07-05	Enrolled
6	306	1	203	2026-07-06	Completed
7	307	6	203	2026-07-07	Enrolled
8	308	7	204	2026-07-08	Completed
9	309	2	204	2026-07-09	Enrolled
10	310	8	205	2026-07-10	Completed
11	311	3	205	2026-07-11	Enrolled
12	312	5	206	2026-07-12	Completed
13	313	6	206	2026-07-13	Enrolled

	assessmentid	enrollmentid	assessmentname	score	assessmentdate
1	401	301	Midterm Exam	88.00	2026-08-01
2	402	301	Final Exam	92.00	2026-09-01
3	403	302	Midterm Exam	76.00	2026-08-01
4	404	302	Final Exam	84.00	2026-09-01
5	405	303	Midterm Exam	81.00	2026-08-02
6	406	304	Midterm Exam	90.00	2026-08-03
7	407	304	Final Exam	94.00	2026-09-03
8	408	305	Midterm Exam	72.00	2026-08-03
9	409	306	Programming Test	95.00	2026-08-04
10	410	306	Final Exam	91.00	2026-09-04
11	411	307	Programming Test	79.00	2026-08-04
12	412	308	AI Assignment	87.00	2026-08-05
13	413	308	Final Exam	93.00	2026-09-05
14	414	309	AI Assignment	74.00	2026-08-05
15	415	310	Web Project	89.00	2026-08-06
16	416	310	Final Exam	91.00	2026-09-06
17	417	311	Web Project	78.00	2026-08-06
18	418	312	DSA Test	86.00	2026-08-07
19	419	312	Final Exam	90.00	2026-09-07
20	420	313	DSA Test	73.00	2026-08-07

10. Normalization

The LearnHub database is designed up to Third Normal Form (3NF).

10.1 First Normal Form (1NF)

All attributes contain atomic values and there are no repeating groups. Each table has a defined primary key and each field stores a single logical value.

10.2 Second Normal Form (2NF)

All non-key attributes depend on the complete primary key. Instructor information is stored separately in the Instructor table and Course stores only InstructorID as the foreign key. This avoids repeatedly storing instructor name and department in every course row.

10.3 Third Normal Form (3NF)

There are no transitive dependencies between non-key attributes. InstructorName and Department depend on InstructorID, while CourseName and Credits depend on CourseID.

Normalized relations:

Student(StudentID, StudentName, Email, Phone)

Instructor(InstructorID, InstructorName, Department)

Course(CourseID, CourseName, InstructorID, Credits)

Enrollment(EnrollmentID, StudentID, CourseID, EnrollDate, CompletionStatus)

Assessment(AssessmentID, EnrollmentID, AssessmentName, Score, AssessmentDate)

11. Required SQL Queries

Query 1 — Display All Students

SQL technique: SELECT * FROM Student;

	studentid	studentname	email	phone
1	1	Aditya Yadav	aditya@example.com	9876543210
2	2	Aarav Sharma	aarav@example.com	9876543211
3	3	Riya Patel	riya@example.com	9876543212
4	4	Kabir Mehta	kabir@example.com	9876543213
5	5	Ananya Joshi	ananya@example.com	9876543214
6	6	Vihaan Shah	vihaan@example.com	9876543215
7	7	Ishita Rao	ishita@example.com	9876543216
8	8	Arjun Nair	arjun@example.com	9876543217

Query 2 — Courses with Instructor Details

SQL technique: JOIN Course with Instructor using InstructorID.

	courseid	coursename	instructorname	department
1	201	Database Management Systems	Dr. Neha Sharma	Computer Science
2	202	Computer Networks	Prof. Rahul Patil	Information Technology
3	203	Java Programming	Dr. Neha Sharma	Computer Science
4	204	Artificial Intelligence	Dr. Sneha Kulkarni	Artificial Intelligence
5	205	Web Development	Prof. Rahul Patil	Information Technology
6	206	Data Structures	Dr. Sneha Kulkarni	Artificial Intelligence

Query 3 — Students Not Enrolled in Instructor 101's Courses

SQL technique: Uses a NOT IN subquery with Enrollment and Course.

	studentid	studentname	email
1	4	Kabir Mehta	kabir@example.com
2	5	Ananya Joshi	ananya@example.com
3	7	Ishita Rao	ishita@example.com
4	8	Arjun Nair	arjun@example.com

Query 4 — Average Assessment Score per Course

SQL technique: Uses JOIN, AVG, ROUND, GROUP BY and ORDER BY.

	courseid	coursename	averagescore
1	201	Database Management Systems	84.20
2	202	Computer Networks	85.33
3	203	Java Programming	88.33
4	204	Artificial Intelligence	84.67
5	205	Web Development	86.00
6	206	Data Structures	83.00

Query 5 — Enrollment Count per Course

SQL technique: Uses LEFT JOIN and COUNT.

	courseid	coursename	enrollmentcount
1	201	Database Management Systems	3
2	202	Computer Networks	2
3	203	Java Programming	2
4	204	Artificial Intelligence	2
5	205	Web Development	2
6	206	Data Structures	2

Query 6 — Completed Enrollments

SQL technique: Filters Enrollment records using CompletionStatus = 'Completed'.

	studentname	coursename	enrolldate
1	Aarav Sharma	Database Management Systems	2026-07-02
2	Aditya Yadav	Database Management Systems	2026-07-01
3	Kabir Mehta	Computer Networks	2026-07-04
4	Aditya Yadav	Java Programming	2026-07-06
5	Ishita Rao	Artificial Intelligence	2026-07-08
6	Arjun Nair	Web Development	2026-07-10
7	Ananya Joshi	Data Structures	2026-07-12

Query 7 — Students with Their Enrolled Courses

SQL technique: Uses JOIN across Student, Enrollment and Course.

	studentname	coursename	completionstatus
1	Aarav Sharma	Artificial Intelligence	Enrolled
2	Aarav Sharma	Database Management Systems	Completed
3	Aditya Yadav	Database Management Systems	Completed
4	Aditya Yadav	Java Programming	Completed
5	Ananya Joshi	Data Structures	Completed
6	Ananya Joshi	Computer Networks	Enrolled
7	Arjun Nair	Web Development	Completed
8	Ishita Rao	Artificial Intelligence	Completed
9	Kabir Mehta	Computer Networks	Completed
10	Riya Patel	Database Management Systems	Enrolled
11	Riya Patel	Web Development	Enrolled
12	Vihaan Shah	Data Structures	Enrolled
13	Vihaan Shah	Java Programming	Enrolled

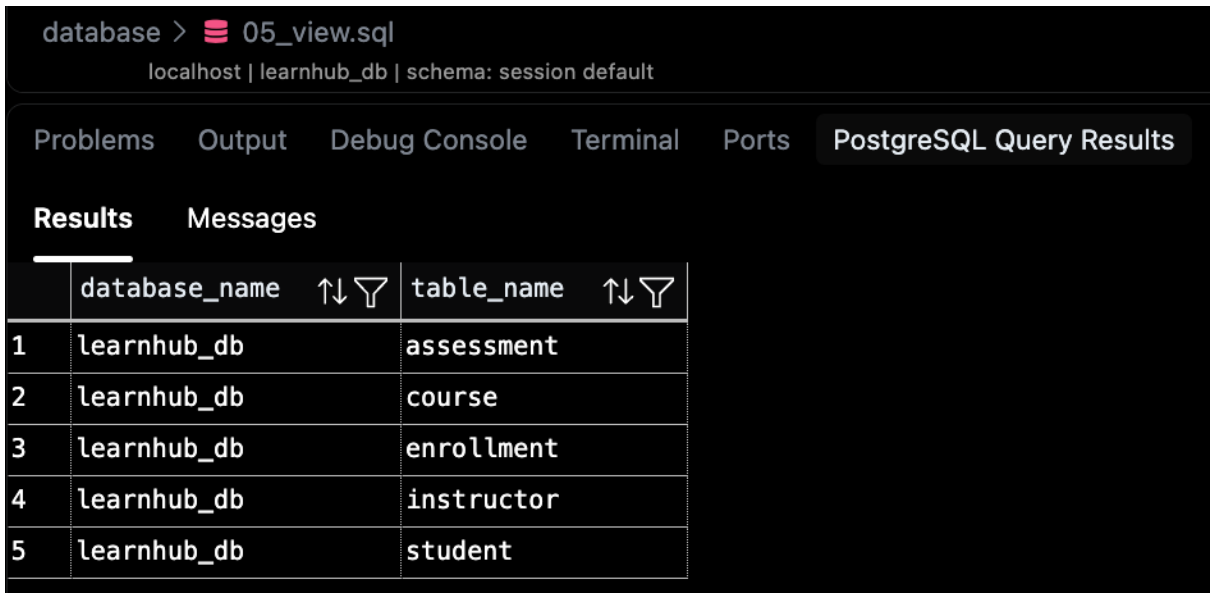
12. InstructorDashboard View

The InstructorDashboard view combines Instructor, Course, Enrollment, and Assessment data. It provides one row per course with instructor details, enrollment count, and average assessment score.
The view uses COUNT(DISTINCT EnrollmentID) for enrollment totals and ROUND(AVG(Score), 2) for average assessment scores.

	instructorid	instructorname	courseid	coursename	enrollmentcount	averagescore
1	101	Dr. Neha Sharma	201	Database Management Systems	3	84.20
2	101	Dr. Neha Sharma	203	Java Programming	2	88.33
3	102	Prof. Rahul Patil	202	Computer Networks	2	85.33
4	102	Prof. Rahul Patil	205	Web Development	2	86.00
5	103	Dr. Sneha Kulkarni	204	Artificial Intelligence	2	84.67
6	103	Dr. Sneha Kulkarni	206	Data Structures	2	83.00

13. Execution and Verification

Database connection established on localhost.
Database learnhub_db created successfully.
Five tables created successfully.
Sample records inserted successfully.
Required SQL queries executed successfully.
InstructorDashboard view created and queried successfully.



	database_name	table_name
1	learnhub_db	assessment
2	learnhub_db	course
3	learnhub_db	enrollment
4	learnhub_db	instructor
5	learnhub_db	student

	instructorid	instructorname	courseid	coursename	enrollmentcount	averagescore
1	101	Dr. Neha Sharma	201	Database Management Systems	3	84.20
2	101	Dr. Neha Sharma	203	Java Programming	2	88.33
3	102	Prof. Rahul Patil	202	Computer Networks	2	85.33
4	102	Prof. Rahul Patil	205	Web Development	2	86.00
5	103	Dr. Sneha Kulkarni	204	Artificial Intelligence	2	84.67
6	103	Dr. Sneha Kulkarni	206	Data Structures	2	83.00

13. Project File Structure

```
LearnHub/
├── database/
│   ├── 01_create_database.sql
│   ├── 02_create_tables.sql
│   ├── 03_insert_data.sql
│   ├── 04_queries.sql
│   └── 05_view.sql
├── diagrams/
│   └── ER Diagram - Learnhub.png
├──documentation/
│   └──  DBMS CASE STUDY - ADITYA YADAV.pdf
│
└──  README.md
```

14. Conclusion

The LearnHub project successfully demonstrates the design and implementation of a relational academic database using PostgreSQL. The system uses primary and foreign keys to maintain relationships, constraints to improve data integrity, SQL queries to retrieve and analyze information, and a database view to provide an instructor-oriented summary. The completed database provides a structured foundation for managing academic learning records.

15. Future Scope

- Add user authentication and role-based access control.
- Create a web-based interface for students and instructors.
- Add attendance and assignment submission modules.
- Add automated reports and graphical analytics.
- Add stored procedures, triggers, and transaction-based workflows.